



Fully Coherent Model of Dynamics around Binary Asteroids: Application to Didymos and the DART Impact

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Abstract

The dynamical environment around binary asteroids has been studied in depth, but these models typically make assumptions about the eccentricity, shapes, or rotational stability of the asteroids. In this work, we develop a fully coherent dynamical model accounting for the gravitational influence of two arbitrarily shaped asteroids and the Sun, allowing for eccentricity and irregular spin states within the binary asteroid. We apply this model to Didymos after the DART impact to determine the stability of orbits in the perturbed system. We then study the behavior of ejecta and identify regions in which ejecta may be long-lived, a concern for the Hera mission. It is likely debris from the DART impact will survive in the system throughout the duration of the Hera mission, and these particles are generally at least 10 cm in diameter and found in wide, eccentric, and inclined orbits. The most likely region for Hera to encounter these particles is just outside the orbit of Dimorphos.

Unified Astronomy Thesaurus concepts: Asteroid dynamics (2210); Asteroid satellites (2207); Ejecta (453); Astrodynamics (76); N-body simulations (1083)

1. Introduction

Binary asteroids, consisting of a primary asteroid orbited by a smaller, synchronously rotating secondary, are common among near-Earth asteroids and have recently been targeted by several missions (P. Pravec et al. 2006). The dynamics within these systems are complex owing to strong spin–orbit coupling caused by the irregular shapes and proximity of the asteroids (D. J. Scheeres 2009). Thus, studying the motion of a particle within these systems is a difficult problem but is of interest to mission designers. Typically, studies make limiting assumptions, such as an equilibrium configuration within the binary asteroid or spherical bodies (J. Bellerose & D. Scheeres 2008; J. A. Arredondo et al. 2012). For a wider applicability to real-world systems, we develop a framework for studying the motion of a particle in the vicinity of binary asteroids, making no assumptions about the system dynamics or asteroid shapes.

The main application of this model is to the Didymos binary asteroid system. On 2022 September 26, the DART spacecraft impacted Dimorphos, the secondary of Didymos, as the first test of a kinetic impactor for the purpose of planetary defense (R. T. Daly et al. 2023). The impact caused a Δv of around 2.7 mm s^{-1} in the orbital velocity of Dimorphos (A. F. Cheng et al. 2023; S. P. Naidu et al. 2024), reducing the mutual orbit period by around 33 minutes (C. A. Thomas et al. 2023). The postimpact orbit now has an eccentricity of around 0.03 (A. J. Meyer et al. 2023; S. P. Naidu et al. 2024).

In 2026, ESA’s Hera mission is planned to rendezvous with Didymos to complete an in-depth study of the effects of the DART impact (P. Michel et al. 2022). There have been a number of studies focused on determining stable orbits around Didymos, which could be leveraged by Hera (L. Dell’Elce et al. 2017; A. Capannolo et al. 2019; I. Fodde et al. 2023). However, while these studies generally include the nonspherical shape of one or

both asteroids, they do not include the eccentricity caused by the DART impact or the perturbed spin of Dimorphos. Another consideration is the fate of ejecta caused by the impact, as ejecta that remains in a stable orbit over long time periods can pose a potential hazard to the spacecraft. Studies have also focused on this problem, but again these have relied on lower-fidelity dynamical models (A. Rossi et al. 2022; K. Langner et al. 2024; G. Madeira et al. 2024; F. Moreno et al. 2024), typically assuming Keplerian dynamics, a circular mutual orbit, a tidally locked Dimorphos, or some combination of these. Understanding the stability of orbits around post-DART Didymos is important for mission planning for Hera as stable orbit regimes are useful for the spacecraft but may accumulate debris for long periods of time, posing a threat to the spacecraft.

Studies of the orbital dynamics around binary asteroids have not included both the eccentricity and spin state of the asteroids as this problem is difficult and computationally intensive. But the influence of these parameters is important as they could change the stability of orbits within the system. Furthermore, Dimorphos is likely attitude-unstable following the DART impact and may be tumbling due to the perturbation from DART (D. C. Richardson et al. 2022; A. J. Meyer et al. 2023; P. Pravec et al. 2024; P. Scheirich et al. 2024). Current models also only use a second-order expansion of the gravitational potential and also employ the MacCullagh approximation of gravity, which is inaccurate when the separation between two bodies is small, as is the case for binary asteroids and close encounters with potential debris in the system. We have developed a fully coherent model of the dynamics around a binary asteroid that incorporates the full spin–orbit coupling within the binary asteroids, the nonspherical gravity terms of the asteroids, and solar effects and can be expanded to an arbitrary order in the gravity field. By “fully coherent,” we mean that the dynamics of the third body are influenced by the dynamics of the full two-body problem (F2BP); these internal dynamics are self-consistent. This model is implemented within the General Use Binary Asteroid Simulator¹ (A. B. Davis & D. J. Scheeres 2020),



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¹ This code is available at <https://github.com/meyeralexj/gubas/tree/USRF4BP>.

and we use this framework to study the stability of orbits around the postimpact Didymos system.

The implementation of this dynamical model is the main contribution of this work. While the model permits the use of any degree-and-order gravity field for the user's choice of fidelity, in this application we will use a fourth degree-and-order gravitational field and model the asteroids as ellipsoids to balance accuracy with computational efficiency. However, any polyhedral shape model could be used, up to any gravitational expansion order. Thus, this model is useful for mission designers concerned with high-fidelity simulations close to a binary asteroid system. In our application to the Didymos system, we will compare our results to previous results using simpler models. As we will see, our results closely match those found in the literature, serving to validate our model and the assumptions made in other studies.

In the remainder of this paper, we first derive the dynamical model of a particle around a binary asteroid in Section 2. This section also includes a discussion of the postimpact dynamics of Didymos. We then provide results of simulations on the stability of orbits around Didymos that may be utilized by the Hera mission in Section 3. In Section 4, we then apply this model to study the dynamics and longevity of debris in the system following the DART impact. Lastly, we provide a discussion in Section 5 and conclude in Section 6.

2. Dynamical Model

We begin by describing our dynamical model used in this work. We break this into two sections. First, we will outline the dynamics governing the binary asteroid system, that is, the F2BP. Then we add the dynamics of a particle under the influence of the binary asteroid system.

2.1. Didymos Dynamics

Because of the proximity and nonspherical shapes of the asteroids within a binary, these systems are governed by spin-orbit coupling, dubbed the F2BP. The most useful forms of the equations of motion of the F2BP are written in the body-fixed frame of the primary asteroid to reduce the total degrees of freedom to the relative problem (A. J. Maciejewski 1995). Thus, we define the primary's body-fixed $\mathcal{A}$ frame aligned with the asteroid's principal moments of inertia, so that its inertia matrix in this frame is diagonal. We also define a similar frame for the secondary, the $\mathcal{B}$ frame, aligned with the secondary's principal moments of inertia. Including the effects of solar gravity and assuming the Sun is well separated from the asteroids (A. J. Meyer & D. J. Scheeres 2021), the equations of motion are written in the $\mathcal{A}$ frame as

$$\ddot{\mathbf{r}} = -\dot{\boldsymbol{\omega}}_A \times \mathbf{r} - 2\boldsymbol{\omega}_A \times \dot{\mathbf{r}} - \boldsymbol{\omega}_A \times \boldsymbol{\omega}_A \times \mathbf{r} - \frac{1}{m} \frac{\partial U_{A/B}}{\partial \mathbf{r}} + \frac{1}{m_B} \frac{\partial U_{S/B}}{\partial \mathbf{r}_{S/B}} - \frac{1}{m_A} \frac{\partial U_{S/A}}{\partial \mathbf{r}_{S/A}}, \quad (1)$$

$$\dot{\mathbf{L}}_A = \mathbf{L}_A \times \boldsymbol{\omega}_A + \mathbf{M}_A + \mathbf{r}_{S/A} \times \frac{\partial U_{S/A}}{\partial \mathbf{r}_{S/A}}, \quad (2)$$

$$\dot{\mathbf{L}}_B = \mathbf{L}_B \times \boldsymbol{\omega}_A + \mathbf{M}_B + \mathbf{r}_{S/B} \times \frac{\partial U_{S/B}}{\partial \mathbf{r}_{S/B}}. \quad (3)$$

Here, $\mathbf{r}$ is the position of the secondary asteroid's barycenter with respect to the primary's barycenter written in the $\mathcal{A}$ frame, $\boldsymbol{\omega}_A$ is the angular velocity of the primary relative to the inertial frame, and $\mathbf{L}_A$ and $\mathbf{L}_B$ are the angular momenta of the primary

and secondary, respectively, again written in the $\mathcal{A}$ frame. $\mathbf{r}_{S/A}$ is the position of the Sun with respect to the primary, and similarly for $\mathbf{r}_{S/B}$ for the secondary. These vectors are also written in the $\mathcal{A}$ frame. We use the reduced mass $m = \frac{m_A m_B}{m_A + m_B}$, and we assume that the mass of the Sun is much larger than the mass of the asteroids. $U_{A/B}$ is the mutual gravitational potential, written for the primary and secondary as

$$U_{A/B} = \int_A \int_B \left(\frac{dm_A dm_B}{|\mathbf{r} - \mathbf{r}_A + \mathbf{C}_A^T \mathbf{C}_B \mathbf{r}_B|} \right). \quad (4)$$

Here, $\mathbf{C}_A$ is the rotation matrix that transforms a vector written in the primary $\mathcal{A}$ frame to the inertial frame, and similarly for $\mathbf{C}_B$ and the secondary $\mathcal{B}$ frame. $\mathbf{r}_A$ and $\mathbf{r}_B$ locate a differential mass element in either body relative to its barycenter, written in each respective body-fixed frame. We can extend this same definition for $U_{S/A}$ and $U_{S/B}$, which are the mutual potential between the asteroids and the Sun. Under the assumption of a spherical Sun, the double integral can be simplified to a single integral over the nonspherical asteroid. $\mathbf{M}_A$ and $\mathbf{M}_B$ are the torques on the primary and secondary caused by the opposite body, written in the $\mathcal{A}$ frame. In Equations (1)–(3), the dot indicates a time derivative taken in the $\mathcal{A}$ frame.

In this formulation, the vectors $\mathbf{r}$, $\mathbf{r}_{S/A}$, and $\mathbf{r}_{S/B}$, along with $\mathbf{L}_A$, $\mathbf{L}_B$, $\mathbf{M}_A$, and $\mathbf{M}_B$, are all written in the $\mathcal{A}$ frame. $\mathbf{r}_A$ and $\mathbf{r}_B$, which locate a differential mass element within each body, are each written in their respective body-fixed frame.

The first step in applying our model to Didymos is to ensure we are accurately replicating the dynamics of the system following the DART impact. Given the high degree of uncertainty surrounding the physical parameters of the system, this is a nontrivial task. We follow the common approach within the literature of calculating the system mass using the preimpact observations, then applying an instantaneous impulse to the secondary to match the system to the postimpact observations (A. F. Cheng et al. 2023; A. J. Meyer et al. 2023; P. Pravec et al. 2024). Thus, there are three key pieces of information provided by observations we will leverage in our model: the preimpact orbit period, the postimpact orbit period, and the postimpact apsidal precession rate.

Given the non-Keplerian dynamics within the Didymos binary asteroid, we define the orbit period as the sidereal period of Dimorphos around Didymos rather than the Keplerian orbit period. This has also been called the stroboscopic orbit period in the literature (A. J. Meyer et al. 2023). In our model, we start with the pre-DART impact system and iterate the bulk density of Didymos until we replicate the preimpact stroboscopic orbit period observed. We initialize the iteration with a guess of $\rho = 2.8 \text{ g cm}^{-3}$ given in S. P. Naidu et al. (2024), and the final result is listed in Table 1. We then iterate a Δv in the orbital velocity, antiparallel to the orbital velocity of Dimorphos, until we replicate the postimpact stroboscopic orbit period. We use the best-fit orbit solution obtained using a combination of photometric and radar observations of Didymos (S. P. Naidu et al. 2024). We use both the derived preimpact shape of Dimorphos (R. T. Daly et al. 2023) to calculate the system mass and the postimpact shape of Dimorphos (S. P. Naidu et al. 2024) to ensure our model produces the correct apsidal precession rate, since this quantity strongly depends on the secondary shape in binary asteroids (M. Čuk & D. Nesvorný 2010; A. J. Meyer & D. J. Scheeres 2024). Thus, we

Table 1
The Parameters of Didymos and Dimorphos Used in Our Simulations

Parameter	Units	Preimpact	Postimpact	Source
Orbit period	hr	11.921 ± 0.000016	11.367 ± 0.0004	S. P. Naidu et al. (2024)
Semimajor axis	m	1189 ± 0.017	1152 ± 0.018	A. J. Meyer et al. (2023)
Preimpact eccentricity	...	0 ± 0.03	0.027 ± 0.0015	S. P. Naidu et al. (2024)
Bulk density	g cm^{-3}	2.794	2.794	This work
Didymos equatorial semiaxis	m	394 ± 11	394 ± 11 m	S. P. Naidu et al. (2024)
Didymos polar semiaxis	m	290 ± 16	290 ± 16	S. P. Naidu et al. (2024)
Dimorphos major semiaxis	m	88.5 ± 2	96 ± 6	R. T. Daly et al. (2023); S. P. Naidu et al. (2024)
Dimorphos intermediate semiaxis	m	87 ± 4	74 ± 4	R. T. Daly et al. (2023); S. P. Naidu et al. (2024)
Dimorphos minor semiaxis	m	58 ± 2	59 ± 7	R. T. Daly et al. (2023); S. P. Naidu et al. (2024)

incorporate the effects of reshaping within the secondary asteroid caused by the impact. In both cases, we use the best-fit ellipsoid, as only half the asteroid was actually observed by the DART spacecraft. These parameters are given in Table 1.

We also include an out-of-plane impulse and torque on Dimorphos to simulate the true impact geometry of DART. These perturbations can produce non-principal-axis rotation in Dimorphos, which was a predicted effect of the DART impact (H. F. Agrusa et al. 2021). The location of the DART impact in the body frame of Dimorphos is given in R. T. Daly et al. (2023),

$$\mathbf{r}_{\text{Imp}} = [-8.31, -83.2, -13]^T \text{ m}, \quad (5)$$

and the relative velocity vector is

$$\mathbf{r}_{\text{Imp}} = [-1.063, 5.964, 1.032]^T \text{ km s}^{-1}, \quad (6)$$

which we use to calculate the torque induced by the impact.

Evidence for the presence of non-principal-axis rotation in Dimorphos has been observed in lightcurves following the impact (P. Pravec et al. 2024). Thus, despite uncertainties within the system, our model faithfully reproduces the best observations of the system, giving the best possible match to reality. For the binary semimajor axis and eccentricity, we adopt the “observable” quantities defined using only the physical separation distance of the two asteroids in order to accurately reproduce observations (A. J. Meyer et al. 2023).

Using our algorithm, we obtain the postimpact dynamics of the Didymos system. The stroboscopic orbit period and separation distance are plotted in Figure 1, showing the evolution of the system. Notably, the orbit period is not constant and experiences fluctuations due to the apsidal precession and angular momentum exchange between the orbit and Dimorphos. This phenomenon is described in detail in A. J. Meyer et al. (2021). We also see fluctuations in the separation distance between the two asteroids. This corresponds to an observable eccentricity of around 0.03. While not a perfect match, these results are generally consistent with the literature (S. P. Naidu et al. 2024) and will serve to examine the dynamics around the system. Our average simulated orbit period is offset from the value reported in the literature by around 30 s, but this offset is smaller than the short-term fluctuations seen in the system and will not appreciably affect any conclusions.

We also examine the attitude evolution of Dimorphos. We track the 1–2–3 Euler angles, corresponding to the roll, pitch, and yaw of Dimorphos relative to the rotating Hill frame. This rotating frame has its x -axis pointing toward Dimorphos and z -axis aligned with the orbital angular momentum. We see these angles evolve over time in Figure 2. The out-of-plane angles,

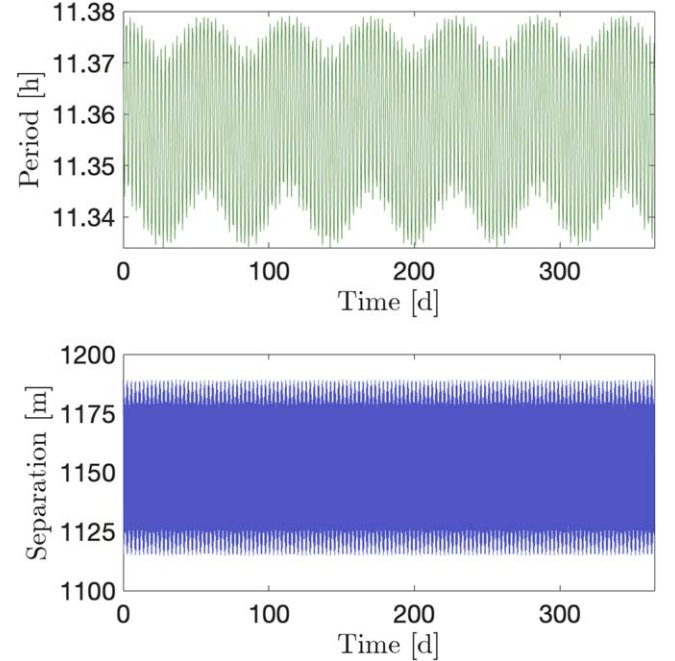


Figure 1. The stroboscopic orbit period (top) and separation distance (bottom) of the Didymos mutual orbit over 1 yr postimpact.

roll and pitch, are nonzero but have small fluctuations. The yaw angle sees large fluctuations with an amplitude of around 30° . Again, these results are consistent with values obtained in S. P. Naidu et al. (2024). However, recent results suggest that the pitch angle may be larger in reality, with an amplitude on the order of 10° (P. Pravec et al. 2024). For now, we continue with these attitude-stable dynamics and discuss the implications of an attitude-unstable Dimorphos later.

2.2. Particle Dynamics

We next define the equations of motion for a test particle under the influence of the two asteroids. We will define the position of the particle as $\boldsymbol{\rho}$ relative to the barycenter of the binary asteroid. $\boldsymbol{\rho}_A$ will denote its position relative to the primary, and similarly for $\boldsymbol{\rho}_B$ relative to the secondary. Still written in the body-fixed frame of the primary, the equations of motion for the particle are

$$\ddot{\boldsymbol{\rho}} = -\dot{\boldsymbol{\omega}}_A \times \boldsymbol{\rho} - 2\boldsymbol{\omega}_A \times \dot{\boldsymbol{\rho}} - \boldsymbol{\omega}_A \times \boldsymbol{\omega}_A \times \boldsymbol{\rho} - \frac{1}{m_P} \left(\frac{\partial U_{P/A}}{\partial \boldsymbol{\rho}_A} + \frac{\partial U_{P/B}}{\partial \boldsymbol{\rho}_B} \right) + \ddot{\mathbf{a}}, \quad (7)$$

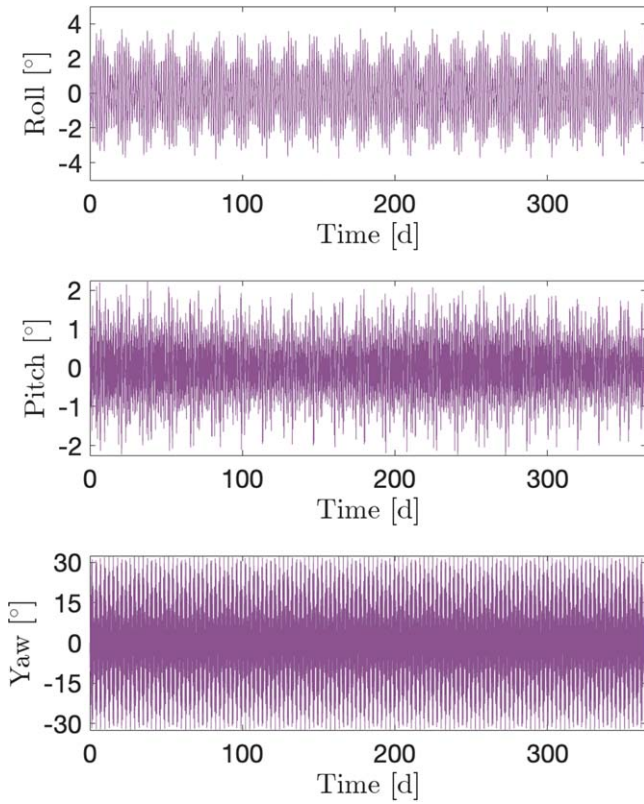


Figure 2. The orbit roll (top), pitch (middle), and yaw (bottom) of Dimorphos relative to the rotating Hill frame over 1 yr.

where the particle has a mass m_p , assumed to be much smaller than the mass of either asteroid, and its mutual potential is defined as $U_{P/A}$ relative to the primary and $U_{P/B}$ relative to the secondary. Using a spherical particle, these can again be reduced to a single integral over the nonspherical asteroid. We have also included $\ddot{\mathbf{a}}$, which is the net acceleration caused by solar effects including both gravity and solar radiation pressure. We assume point-mass attraction between the particle and the Sun, and we use a cannonball model for solar radiation pressure. We also incorporate shadowing, using bounding spheres around the asteroids to approximate shadowing for simplicity. Under these assumptions, the solar acceleration is equal to

$$\ddot{\mathbf{a}} = \frac{Gm_S}{|\mathbf{r}_{S/P}|^2} \hat{\mathbf{r}}_{S/P} - sF_S \frac{C_R A_P}{m_p r_{S/P}^2} \hat{\mathbf{r}}_{S/P}, \quad (8)$$

where $\mathbf{r}_{S/P}$ is the position of the Sun relative to the particle, and the hat corresponds to the unit vector in this direction. $\bar{r}_{S/P}$ is the magnitude of this distance in astronomical units; G is the universal gravitational constant; m_S is the mass of the Sun; F_S is the solar constant, equal to about $1 \times 10^8 \text{ kg km}^3 \text{ s}^{-2} \text{ m}^{-2}$; C_R is the coefficient of reflectivity of the particle; and A_P is the projected area of the particle, which is a function of particle radius. s is our shadow function, equal to 0 if the particle is eclipsed by either asteroid and equal to 1 otherwise.

A diagram of our dynamical system, showing the relevant bodies and vectors used in the equations of motion, is shown in Figure 3. We have developed a fully self-consistent model for simulating the dynamics of a particle under the influence of two

arbitrary rigid bodies and the Sun. The only assumption made in this model is that the masses of the asteroids are much smaller than that of the Sun but much larger than the particle. We make no assumption about the spin or eccentricity of the asteroids. In integrating the equations of motion, we implement a four-step Adams predictor–corrector method, utilizing a two-step Adams–Bashforth predictor and a two-step Adams–Moulton corrector. We use a simple Runge–Kutta to obtain the first four values to begin the integration. To evaluate the mutual potentials and their partial derivatives, we use a recursive algorithm leveraging the inertia integrals truncated at the fourth degree and order (P. Tricarico 2008; X. Hou et al. 2017). The inertia integrals are an alternative to the classical spherical harmonic representation of gravity fields.

3. Hera Spacecraft Simulation Results

In analyzing stable orbits for Hera, we use previous results to inform guesses for initial conditions, then refine those initial conditions in a simple analytic model before simulating the orbits in our high-fidelity framework (L. Dell’Elce et al. 2017). For the spacecraft, we use a mass of 1000 kg, an average projected area of 10 m^2 , and a coefficient of reflectivity of 1.2.

There are five classes of orbits we will study, which were found to be stable in the absence of perturbations in the preimpact Didymos system (L. Dell’Elce et al. 2017). These include direct and retrograde interior (within the orbit of Dimorphos), direct and retrograde exterior (outside the orbit of Dimorphos), and circumsecondary orbits, which orbit around Dimorphos in the rotating frame aligned with Dimorphos. We will also study the L4 equilibrium point. For the L4 point, we place the spacecraft in the vicinity of where the L4 point would be if the two asteroids were point masses in a circular orbit at the same semimajor axis.

We plot these six examples in Figure 4. We see that the interior orbits, both direct and retrograde, remain robust despite the perturbations within the systems, and the spacecraft remains in this region for at least 2 yr. On the other hand, the exterior orbits, both direct and retrograde, become unstable and are not robust to perturbations. In these cases, the spacecraft collides with the primary. For the circumsecondary and L4 point orbits, the orbits are also not robust and impact the secondary. This behavior is expected, since previous analyses have shown that the exterior, circumsecondary, and L4 orbits are generally not robust to perturbations to the system or the spacecraft’s initial velocity (L. Dell’Elce et al. 2017). However, we see that the interior orbits remain remarkably stable over several years despite the perturbations to the system. These consistent results serve to validate our dynamical model.

3.1. Terminator Orbits

Next we examine terminator orbits, which are particularly useful when dealing with solar radiation pressure. These orbits have their angular momentum vector pointing toward the Sun and their eccentricity vector pointing above the ecliptic. The rotation of the longitude of the ascending node is then driven by solar radiation pressure to keep the orbit aligned in the solar direction (D. J. Scheeres 2016).

These orbits provide numerous advantages in addition to stability within an environment with significant solar radiation pressure. The geometry of these orbits means the spacecraft will not be eclipsed and has constant access to solar power.

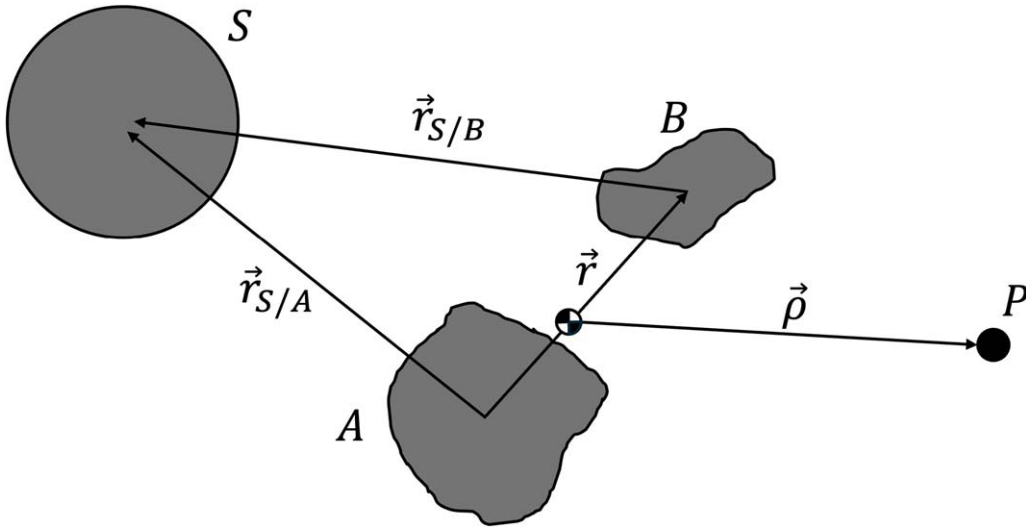


Figure 3. Diagram of the dynamical system, showing the binary asteroid, Sun, and particle, along with the relevant vectors.

Additionally, terminator orbits offer full surface coverage of the asteroid for scientific study. Because of these advantages, terminator orbits have been used in previous small-body missions (S. G. Hesar et al. 2017) and are planned for the Hera mission (A. Felin et al. 2024).

A terminator orbit is robust to solar radiation pressure when the eccentricity is defined as (D. J. Scheeres 2012)

$$e = \cos\Lambda, \quad (9)$$

$$\Lambda = \arctan\left(\frac{3}{2}F_S \frac{C_R A_P}{m_P} \sqrt{\frac{a}{\mu\mu_S a_S(1-e_S^2)}}\right), \quad (10)$$

where μ_S is the standard gravitational parameter of the Sun and a_S and e_S are the heliocentric semimajor axis and eccentricity of Didymos, respectively. Here, μ is the gravitational parameter of the combined system, and a is the semimajor axis of the spacecraft about the binary's barycenter.

Using this approach, we plot two examples of terminator orbits in Figure 5. One of these orbits remains stable, and the other is unstable. The unstable orbit is influenced by a 2:1 resonance with Dimorphos, driving up the eccentricity until the spacecraft impacts with Didymos. These results are consistent with analyses of the preimpact Didymos system (L. Dell'Elce et al. 2017). This indicates that the perturbed system does not strongly change the dynamics of the terminator orbits. Thus, Hera could make use of stable terminator orbits, as long as it is well separated from the 2:1 resonance.

3.2. Tumbling

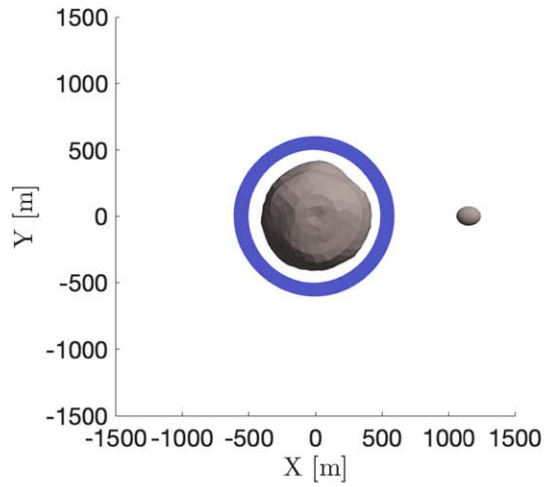
Recent results suggest that Dimorphos is attitude-unstable and experiencing significant out-of-plane rotation postimpact (P. Pravec et al. 2024). It has also been shown in the literature that a tumbling secondary results in an exchange of angular momentum between the secondary and the orbit, resulting in a fluctuation of the orbit's eccentricity on relatively fast timescales (A. J. Meyer et al. 2023). These eccentricity fluctuations could prove to be important in the stability of these orbits. To investigate this, we simulate another direct interior orbit that interacts with a tumbling Dimorphos. The result is shown in Figure 6, where we plot the eccentricities of the binary mutual orbit and the spacecraft's trajectory. This

clearly shows significant fluctuations in the eccentricity of Dimorphos. The spacecraft orbit is close enough to Dimorphos such that it experiences significant perturbations from the evolving eccentricity. Eventually, the spacecraft comes close enough to Dimorphos where its orbit is strongly perturbed, increasing the eccentricity and resulting in an impact onto the primary.

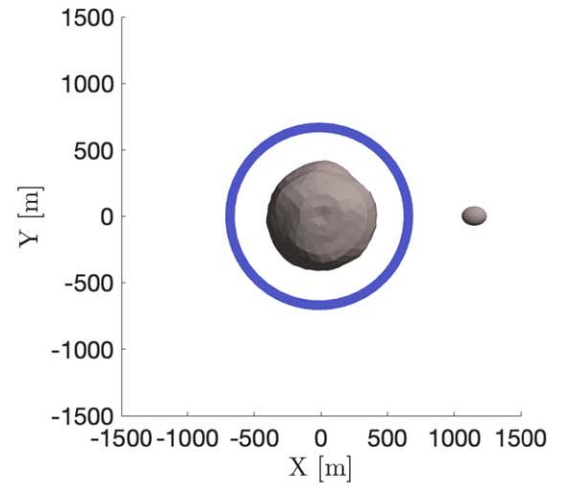
Thus, tumbling in the secondary could potentially affect the stability of orbits that pass close to the moonlet. Some of these trajectories may be stable in the attitude-stable case, but rapid changes to the binary mutual orbit caused by tumbling can drive these trajectories to become unstable.

4. Ejecta Particle Simulation Results

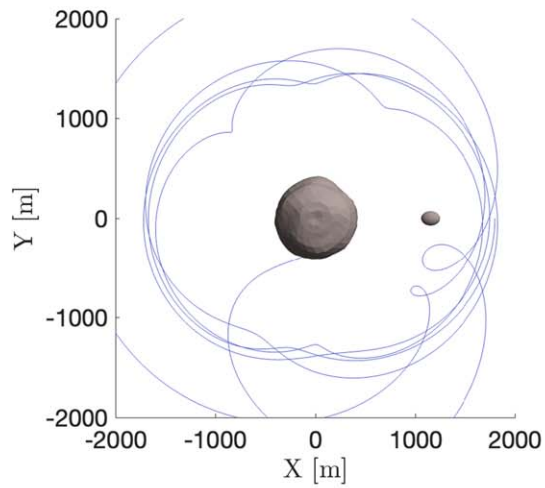
Next we examine the fate of ejecta from the impact in the post-DART system. This analysis is similar to previous works (K. Langner et al. 2024), but we extend the literature by leveraging our newly derived high-fidelity model and examining the predicted location of debris upon Hera's arrival at the Didymos system. We also extend our analysis to smaller particles, where the influence of solar radiation pressure becomes more important. For this study, we will perform a Monte Carlo analysis over possible ejecta physical and dynamical properties. These include the ejection speed and direction and the particle size. We simulate around 6000 particles, evenly sampling over particle radius. We model the particle as a sphere with a radius between 1 cm and 1 m while keeping a constant density of the particle of 3.3 g cm^{-3} , consistent with the expected grain density of Dimorphos (S. Raducan et al. 2024). Thus, the radius of the particle controls both its mass and projected area. This range of particle sizes was selected to examine the dynamics of particles that may pose a risk to Hera. We vary the speed linearly from 6 to 18 cm s^{-1} relative to Dimorphos and vary the direction uniformly anywhere within a cone with a half-angle of 70° , centered on the azimuth at the impact site (A. F. Cheng et al. 2023; J. Deshapriya et al. 2023). The lower bound for ejecta velocity was selected as just below the escape velocity of Dimorphos. The upper bound was determined empirically, as particles with an initial velocity larger than 18 cm s^{-1} are typically on highly eccentric orbits that eventually escape from



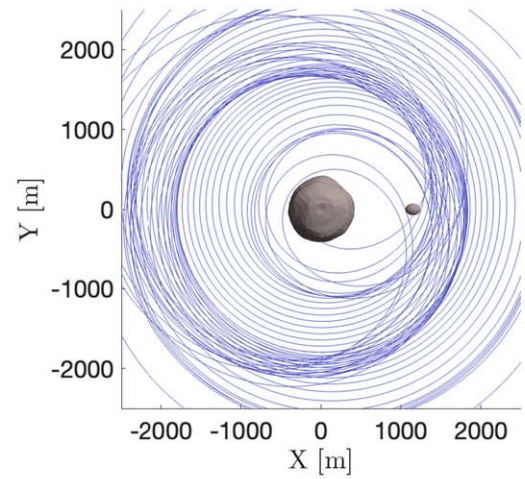
(a) Direct Interior



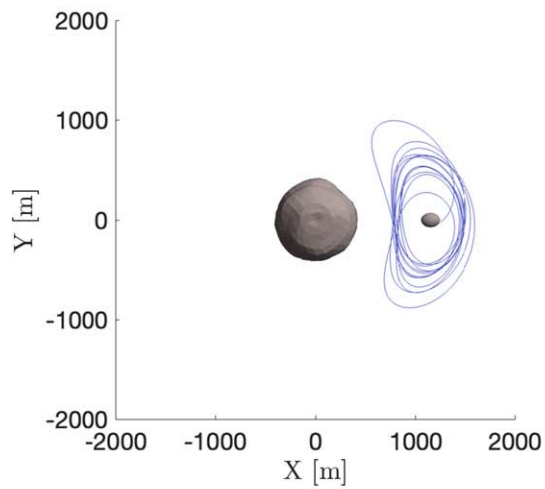
(b) Retrograde Interior



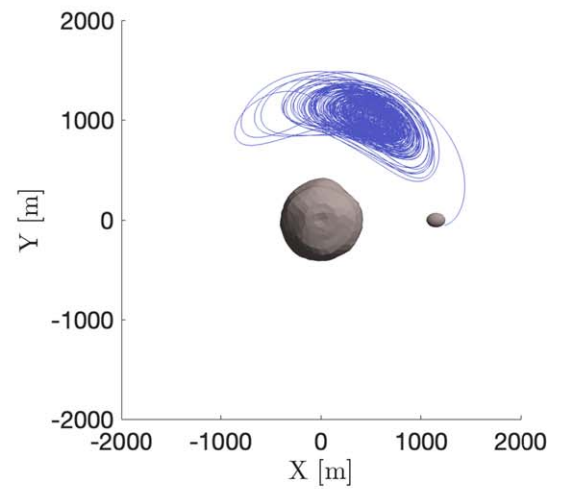
(c) Direct Exterior



(d) Retrograde Exterior



(e) Circumsecondary



(f) L4 Point

Figure 4. Plots of examples for (a) direct interior, (b) retrograde interior, (c) direct exterior, (d) retrograde exterior, (e) circumsecondary, and (f) L4 point orbits. Only the interior orbits remain stable in the perturbed system.

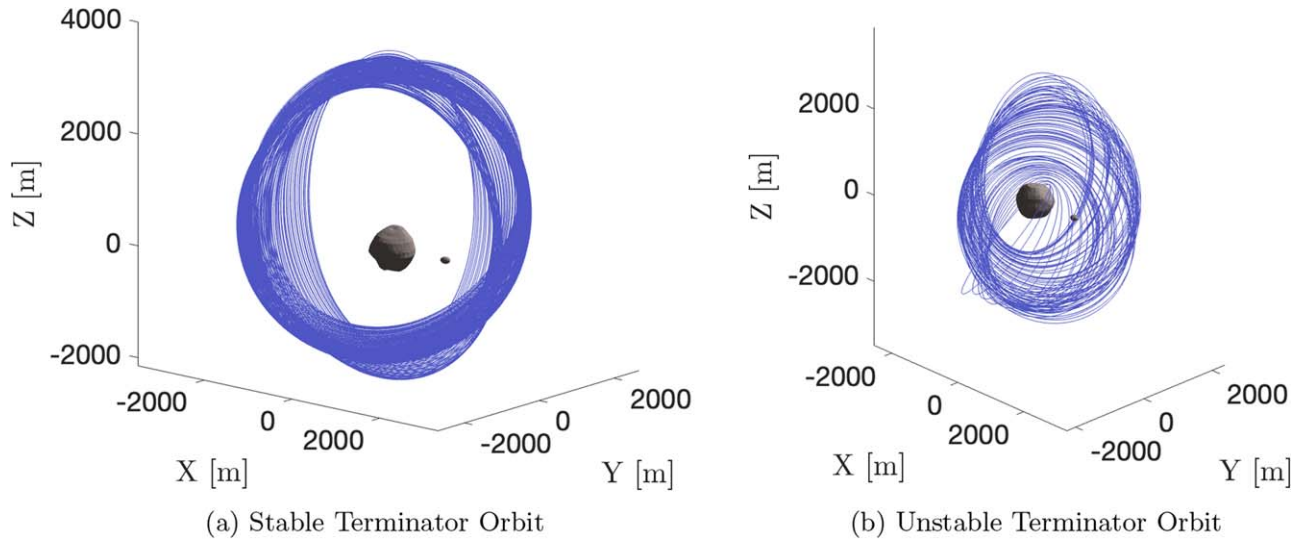


Figure 5. Terminator orbits shown in the rotating Sun frame, showing (a) stable and (b) unstable cases. The unstable orbit is in a 2:1 resonance with Dimorphos.

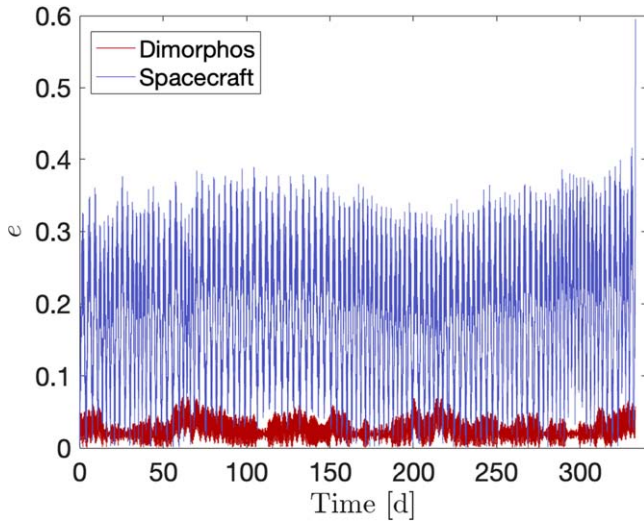


Figure 6. The eccentricities of Dimorphos and the spacecraft trajectory over time. We see that fluctuations in the binary mutual orbit eccentricity also manifest as a strong perturbation on the spacecraft, eventually pushing the spacecraft onto an unstable orbit, resulting in an impact onto the primary.

the system due to solar radiation pressure. We use a coefficient of reflectivity of 1.2 and keep the same binary asteroid parameters as reported in Table 1.

We propagate the particle forward in time until it either impacts Didymos or Dimorphos or escapes the system. Here, escape is defined as when the particle reaches 150 km from Didymos, which is approximately the Hill radius at aphelion. We end our simulations after the particle escapes the system, impacts onto the primary or secondary, or survives 5 yr, as any particles surviving that long may interact with Hera, which will arrive at Didymos about 4 yr after the DART impact and operate for approximately 6 months.

We plot the duration the particle survived as a function of its initial velocity off Dimorphos in Figure 7. This figure is in good agreement with previous analyses (K. Langner et al. 2024), where we see clear populations for quickly reimpacting the secondary for low ejecta velocities and escaping the system for high ejecta velocities. There is also a range of ejecta speeds, between 7 and 17 cm s^{-1} relative to Dimorphos, for which

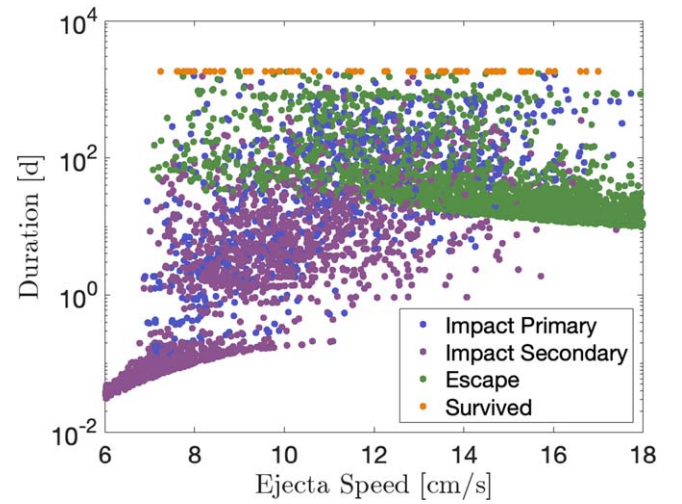


Figure 7. Scatter plot of the duration of each ejecta particle as a function of its initial velocity relative to Dimorphos. We group the simulations into four categories depending on their ultimate fate.

particles can survive long term. In our simulation setup, about 2% of the ejecta particles survive the full 5 yr, which is consistent with past estimates for similar initial ejection velocities (K. Langner et al. 2024; F. Moreno et al. 2024). About 18% collide with the primary, 39% reimpact into the secondary, and 41% escape the system.

Of interest is the properties of long-lasting orbits. To that end, we extend our analysis to calculate the time-averaged inclination, eccentricity, and semimajor axis of the particles and plot these in Figure 8. The surviving ejecta typically have large eccentricities and inclinations, with a semimajor axis on the order of tens of kilometers. These results are again consistent with findings by K. Langner et al. (2024).

In Figure 8, we also plot the ejecta surviving time and eccentricity as a function of the particle radius. From these plots, we see that only particles larger than roughly 5 cm survive until the arrival of Hera. For smaller particles, we see increased eccentricities resulting in escape from the system. This is driven by the larger effect of solar radiation pressure (SRP) on these smaller particles.

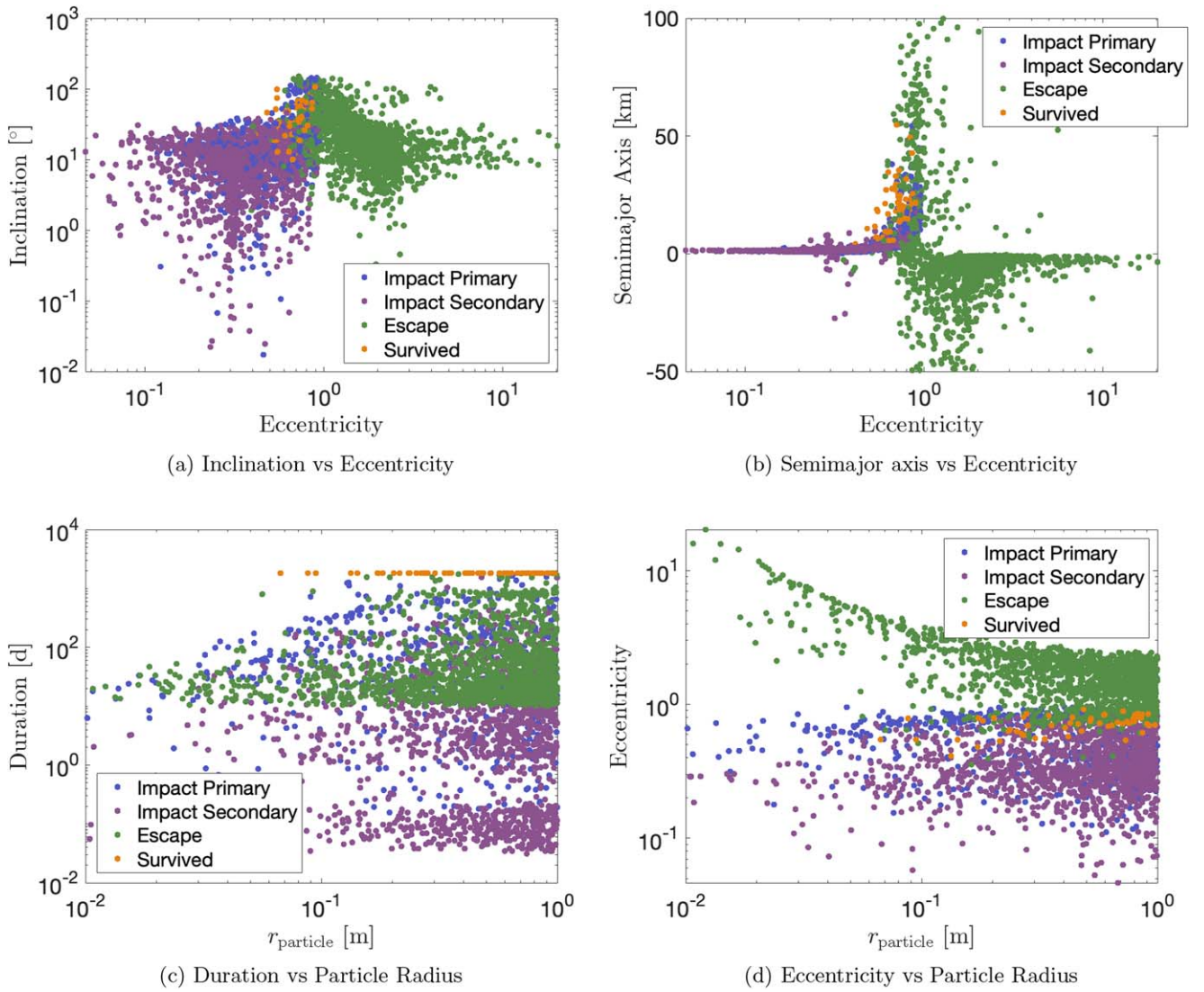


Figure 8. Scatter plots of the time-averaged (a) inclination as a function of eccentricity and (b) semimajor axis as a function of eccentricity for the debris simulations. The duration of each ejecta particle as a function of particle radius (c) and the time-averaged eccentricity as a function of particle radius (d).

To investigate the evolution of the debris over time, we calculate residence time plots in position space, amalgamated over all our simulations. Using 10×10 m bins, each time an ejected particle enters a bin, we add to that bin's total count and continue counting through all simulations. We break this process up into four distinct time periods: up to 1 week after the impact, from 1 week to 1 yr after the impact, from 1 to 4 yr after the impact, and longer than 4 yr after the impact. This last time period illustrates where Hera should expect to find debris. The evolution is shown in Figure 9. At first, we have a high density around the secondary as a large fraction of particles rapidly reimpact the secondary. As time passes, the high-density region expands as particles have time to evolve around the system. We see a concentration of particles centered around the orbit of Dimorphos. Then, moving further away from the impact, the density of particles decreases as some are removed from the system through impacts with the two asteroids or leaving on a heliocentric orbit. At the time of Hera's arrival at Didymos, we expect the highest density of particles to be just outside the orbit of Dimorphos, again consistent with previous studies (F. Moreno et al. 2024).

To show this in greater detail, we plot an example of a long-lived ejecta particle's trajectory in Figure 10. This particle has a radius of 70 cm and an initial ejection speed of 7.8 cm s^{-1} relative to Dimorphos. The plot shows it entering a wide, eccentric orbit around the system. We also plot this in eccentricity and inclination space in Figure 11. After ejection, the particle is in a highly eccentric and inclined orbit for a few years after the impact. Furthermore, we see a cycling between eccentricity and inclination, similar to the Kozai–Lidov effect, which was also identified by K. Langner et al. (2024). This matches our expectations from interpretation of Figures 8 and 9.

As another example, we show a particle that is trapped into a 4:1 resonance with Dimorphos. Its trajectory, in the frame rotating with Dimorphos, is shown in Figure 12. It remains in this resonance for around 2 yr before being perturbed out of the resonance and escaping the system. We plot the orbit period in Figure 13, showing the instantaneous Keplerian orbit period of the particle along with the 4:1 resonance with the stroboscopic orbit period of Dimorphos. This clearly demonstrates the resonance driving this particle. Figure 14 shows the eccentricity and inclination of this particle. The effect of the resonance is

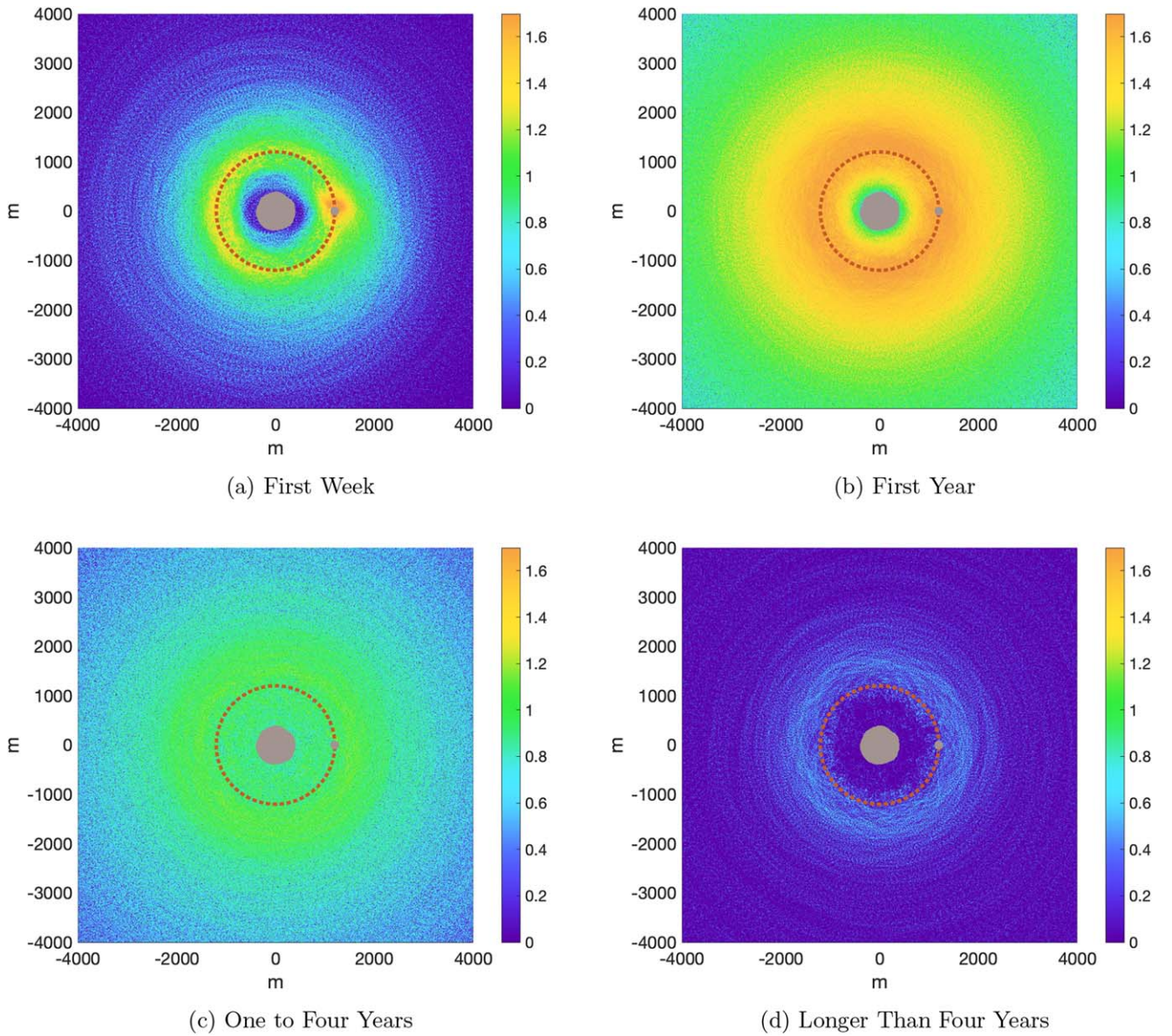


Figure 9. Density heat maps showing where debris is typically found over (a) 1 week, (b) 1 yr, (c) 1–4 yr, and (d) longer than 4 yr after the DART impact. Blue indicates no debris, and orange is a high density of debris. The color scale is logarithmic and truncated at a maximum of 50 particles to aid interpretation.

apparent, where the eccentricity remains nearly constant and the inclination sees periodic variation during this period. Within this study, this is the only example of such a resonance. While outside the scope of this analysis, further study may reveal the importance of resonances within the system and their effect on the fate of ejecta. However, given the rarity of this occurrence in our simulations, it is possible that resonances have a negligible effect when considering the ejecta dynamics as a whole. Nonetheless, we see that it is possible following the DART impact.

4.1. Cumulative Distribution

We next examine the cumulative time distribution of debris following the DART impact. This is shown in Figure 15, where we plot the percent of particles remaining as a function of time. This demonstrates that over 90% of debris particles have left the system within 1 yr after the impact. The distribution begins to level off around 3 yr after the impact. If a debris particle lasts

3 yr after the impact, it has about an 84% chance of lasting an additional year and a 64% chance of lasting the full 5 yr simulation time.

Interestingly, we see two additional periods in which the number of particles leaving the system increases. These occur about 2.2 and 4.3 yr after the impact, aligning with perihelion passages of Didymos. This behavior matches expectations based on similar analyses of DART debris (K. Langner et al. 2024). The initial rapid decrease in debris particles is also exacerbated by a perihelion passage that occurred around 30 days after the DART impact. This pattern demonstrates the influence of the Sun on the dynamics around the system, and we see that perihelion passages are efficient at removing debris.

4.2. Ejecta Velocities

Now that we have established the likely presence of debris in the Didymos system upon Hera’s arrival, the next question to ask is how much of a risk does this debris pose. To answer that

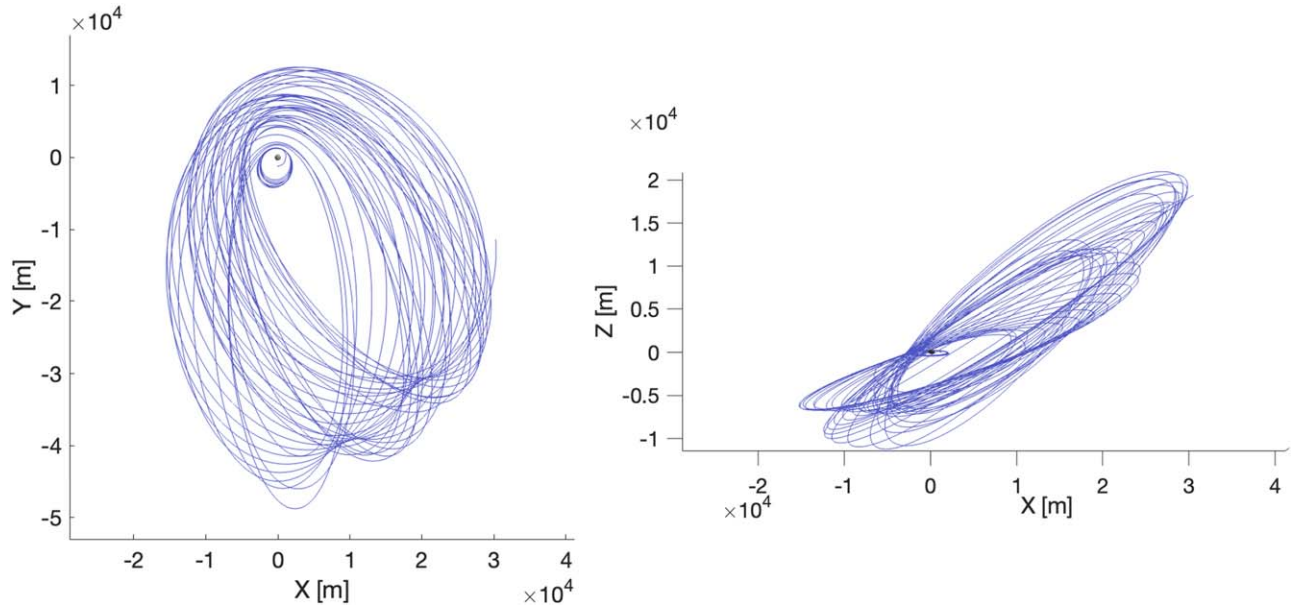


Figure 10. The trajectory of a 70 cm radius debris particle with an initial ejection velocity of 7.8 cm s^{-1} surviving for the full 5 yr, plotted in an inertial frame with the x -axis aligned with the initial position of Dimorphos and the z -axis aligned with the system's total angular momentum.

question, we examine the average periapsis velocity and the size of particles lasting at least 4 yr after the DART impact. These velocities are calculated using the average semimajor axes and eccentricities of the orbits. The longest-lived particles are plotted in Figure 16. We see average periapsis velocities from 5 to 20 cm s^{-1} spanning particle radii from 0.1 to 1 m . Thus, we see that the surviving particles are all roughly larger than 10 cm . This is unsurprising, given the increasing influence of SRP as particle size decreases. We also plot the maximum periapsis velocity experienced by these particles to demonstrate the variability of periapsis velocities and provide an upper bound.

There are a few points to consider from these results. First, while a 0.1 m particle reaching a maximum velocity of 6 cm s^{-1} may not necessarily pose much danger to the spacecraft, a 1 m boulder moving at 15 cm s^{-1} could. This danger is enhanced when we realize that particles on retrograde orbits are possible due to the scattering effect of Dimorphos. Thus, the velocity relative to the spacecraft could potentially be 10 – 20 cm s^{-1} higher. Lastly, we see that the maximum velocities can reach nearly 50 cm s^{-1} . Especially for larger particles, this could potentially pose a hazard to the spacecraft.

4.3. Ejecta Fate

Finally, we examine the fate of the ejecta. Namely, what is the impact flux onto the primary and secondary over time? To answer this question, we plot the number of impacts onto Didymos and Dimorphos over time in Figure 17, along with the escape flux. This shows, in a histogram form, the number of impacts or escapes within a time interval. We see that reimpacting the secondary is common but quickly drops off, with nearly all impacts onto Dimorphos occurring within 1 yr of the DART impact. Impacts onto the primary occur over a longer time period, with the flux dropping off about 3 yr after the DART impact. We also see a similar distribution of ejecta escaping the system, with a secondary peak coinciding with Didymos undergoing its second perihelion passage. After about 3 yr after the DART impact, ejecta is much more rarely

removed from the system, and it most likely escapes the system or impacts Didymos.

This information suggests that upon Hera's arrival at the Didymos system, any secondary craters on Dimorphos are likely a few years old. Newer craters on Didymos are possible, and Hera may even observe such impacts during its study of the system.

5. Discussion

By developing a high-fidelity model for the dynamics of a particle within a perturbed binary asteroid system, we are able to study the stability of orbits around these systems and the evolution of debris created by the DART impact. We incorporate the full spin-orbit coupling between the asteroids within a fourth degree-and-order gravity field. This allows us to include the effects of eccentricity and the attitude instability of Dimorphos. While previous studies have begun to analyze this problem, our framework provides the highest level of accuracy within the literature, albeit at the cost of less computational efficiency. The results obtained using this high-fidelity model are consistent with the current literature, demonstrating the validity of the assumptions used in previous work (L. Dell'Elce et al. 2017; K. Langner et al. 2024; F. Moreno et al. 2024).

We find that both direct and retrograde interior orbits are stable over long time periods, provided they are well separated from the secondary. These orbits could be leveraged by the Hera mission for proximity operations around both Didymos and Dimorphos. However, circumsecondary orbits, which are particularly useful for studying the secondary and are stable in the circular problem, become unstable when eccentricity is included. These orbits could still be used over short time periods, for example, to ultimately land on Dimorphos as planned by one of the cubesats in the Hera mission (P. Michel et al. 2022). Indeed, placing the cubesat in an unstable circumsecondary orbit may be one strategy to safely deliver the spacecraft to the surface, as the relative velocity remains on the order of 10 cm s^{-1} . Exterior orbits quickly become unstable and

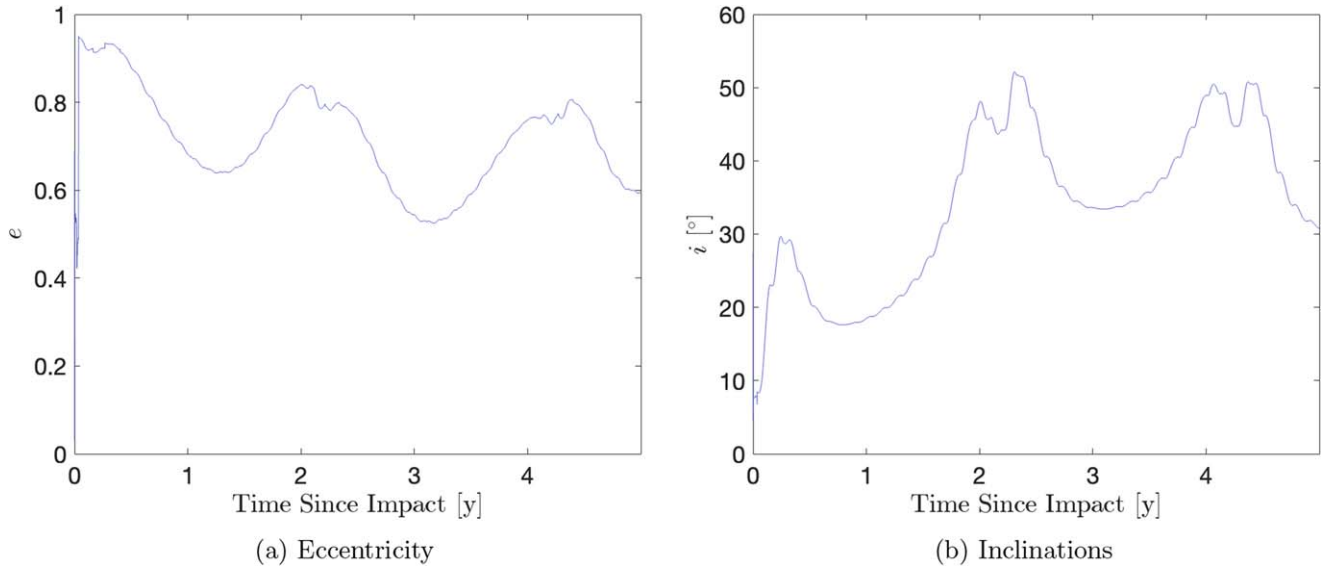


Figure 11. For a long-surviving debris particle with 70 cm radius and an initial ejection velocity of 7.8 cm s^{-1} , plots of (a) the eccentricity and (b) the inclination. This demonstrates a wide, eccentric, and inclined orbital history.

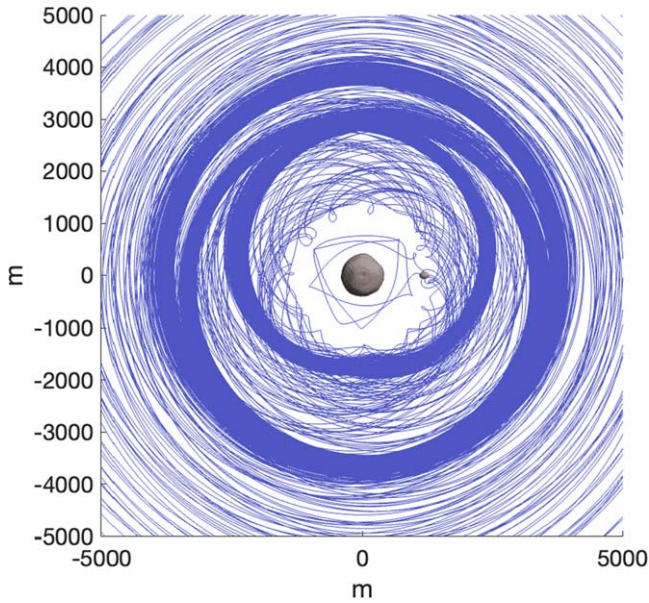


Figure 12. A particle entering a long-term 1:3 resonance with Dimorphos. Eventually, the particle is perturbed out of this resonance before escaping the system.

are not recommended for use, especially as this is the area with the highest population of debris.

Terminator orbits remain stable outside of resonances with Dimorphos. These orbits are also scientifically advantageous for the study of Didymos. Their geometry also limits their exposure to debris still in the system, although the presence of highly inclined debris is a possibility upon Hera's arrival at the system.

As already discussed, one major risk to Hera is the possibility of collisions with debris from the DART impact that has remained stable within the system. We simulated a variety of potentially hazardous particles to test their stability and dynamics. The evolution of ejecta is chaotic and dominated by close encounters with Dimorphos. We find it is likely that some debris will still be present in the system at the time of

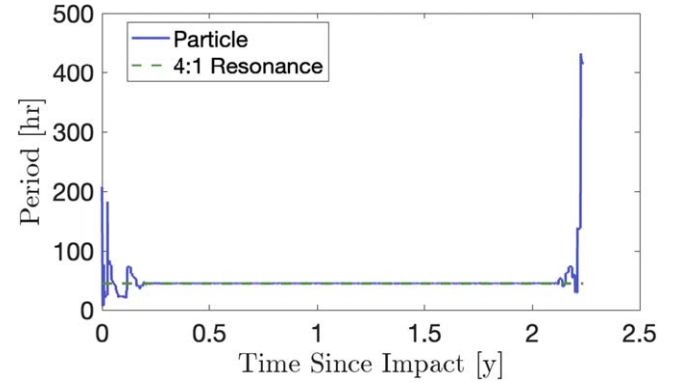


Figure 13. The instantaneous Keplerian orbit period of the particle, along with the 4:1 resonance with the stroboscopic orbit period of Dimorphos.

Hera's arrival. These long-lived particles are all larger than 10 cm in radius and typically found on wide, eccentric, and inclined orbits. One benefit of this is that the stable interior orbits are likely relatively clear of debris, as the highest concentration of particles is centered outside the orbit of Dimorphos. One factor to keep in mind, however, is that any particle that does venture into the interior space will have a higher velocity relative to those remaining in the exterior space. Overall, particles may present a danger to the spacecraft as their high eccentricity and inclination mean they may be difficult for the spacecraft to detect. Furthermore, the orbital velocities within the Didymos system can reach several tens of cm s^{-1} , even for meter-sized boulders. However, given that only around 2% of our particles survive for longer than 4 yr, the probability of a significant collision may be small.

The analysis done in this work primarily uses an attitude-stable Dimorphos. However, evidence suggests that Dimorphos may be attitude-unstable in reality (P. Pravec et al. 2024). In this analysis, we see that some orbits coming close to Dimorphos, such as circumsecondary, are already unstable. Longer-term trajectories, for example, terminator orbits or long-lived debris, do not spend much time near Dimorphos. However, some orbits that are stable and frequently pass close to Dimorphos can be driven unstable by tumbling in the

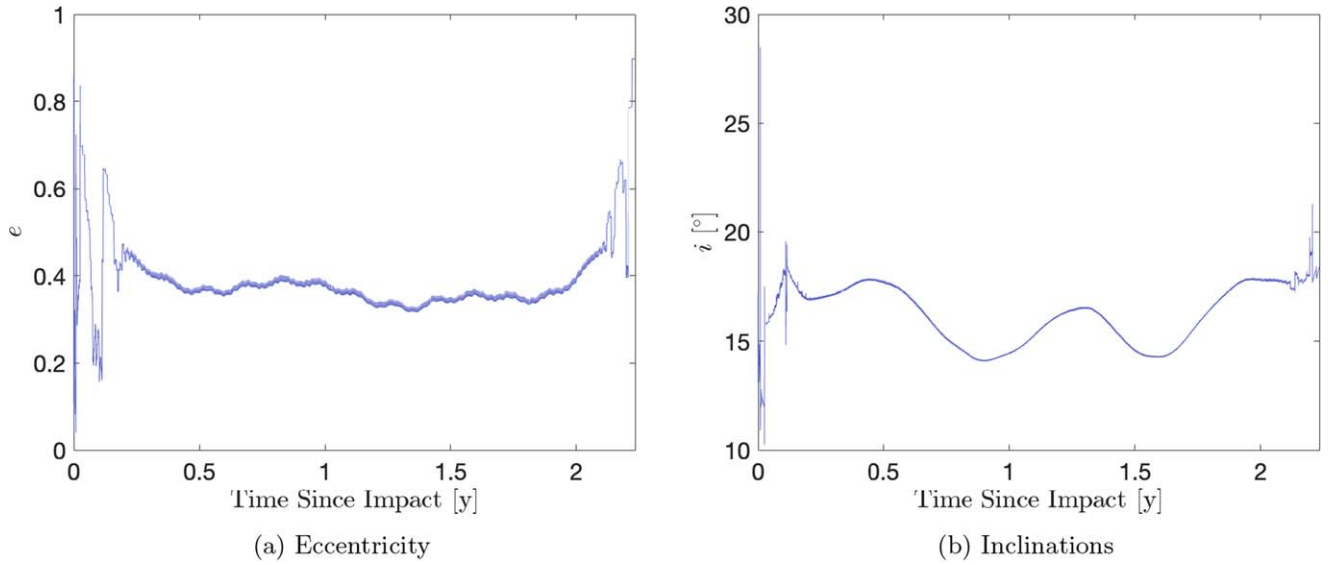


Figure 14. For the 4:1 resonant particle, the (a) eccentricity and (b) inclination of the particle over time. The effect of the resonance on eccentricity is apparent.

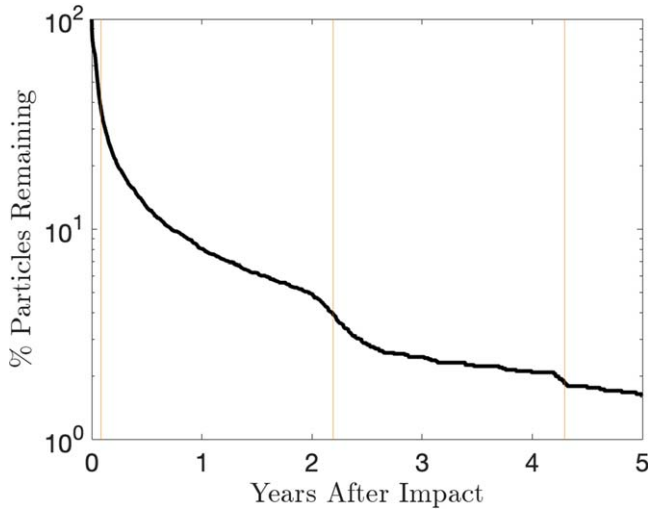


Figure 15. The cumulative distribution of debris, expressed as the percent of remaining debris particles over time. Vertical lines correspond to perihelion passages of the Didymos system.

secondary. As an example, we showed a direct interior orbit that survives for almost a full year before the attitude instability eventually causes a close encounter between the spacecraft and the secondary, forcing a collision into the primary. Thus, to minimize risk to a spacecraft, orbits should be designed to stay well separated from Dimorphos until a thorough understanding of its rotational state and orbit evolution is achieved.

This high-fidelity model can be leveraged to study the motion close to either component of a binary asteroid. In particular, the nonspherical gravity and rotation of the secondary component are important characteristics to consider while planning proximity operations near this body. As the Hera mission is planning such proximity operations around Didymos, this model could be implemented for a more accurate prediction of the spacecraft dynamics. Future work can focus on this analysis. Furthermore, the Hera spacecraft will provide accurate shape models and density profiles of both asteroids that can be leveraged in a similar analysis to more accurately

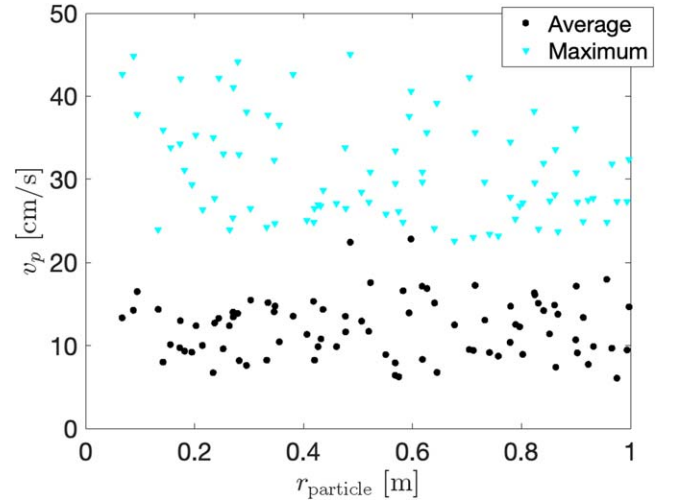


Figure 16. For particles lasting at least 4 yr, the average and maximum periapsis velocity as a function of particle radius.

study the orbital dynamics of particles around the Didymos system.

The primary objective of this paper is to introduce this high-fidelity modeling framework. We perform some analyses to validate this model and compare to similar studies within the literature. Given the similarities between our results and those produced by simplified, but more efficient, models in simulating debris dynamics around the system, the high-fidelity framework used in this work is not as effective at studying this problem. Simpler models are able to simulate a much larger number of particles in order to make stronger conclusions about the ejecta dynamics, whereas the model introduced in this work is not computationally efficient enough to reach those same numbers. However, we have used our high-fidelity model to produce consistent results, helping to validate the assumptions used in simpler models. Instead, we conclude that the benefit of leveraging our high-fidelity framework is in studying and planning the dynamics of a spacecraft within the system, especially near either of the asteroids. As already discussed,

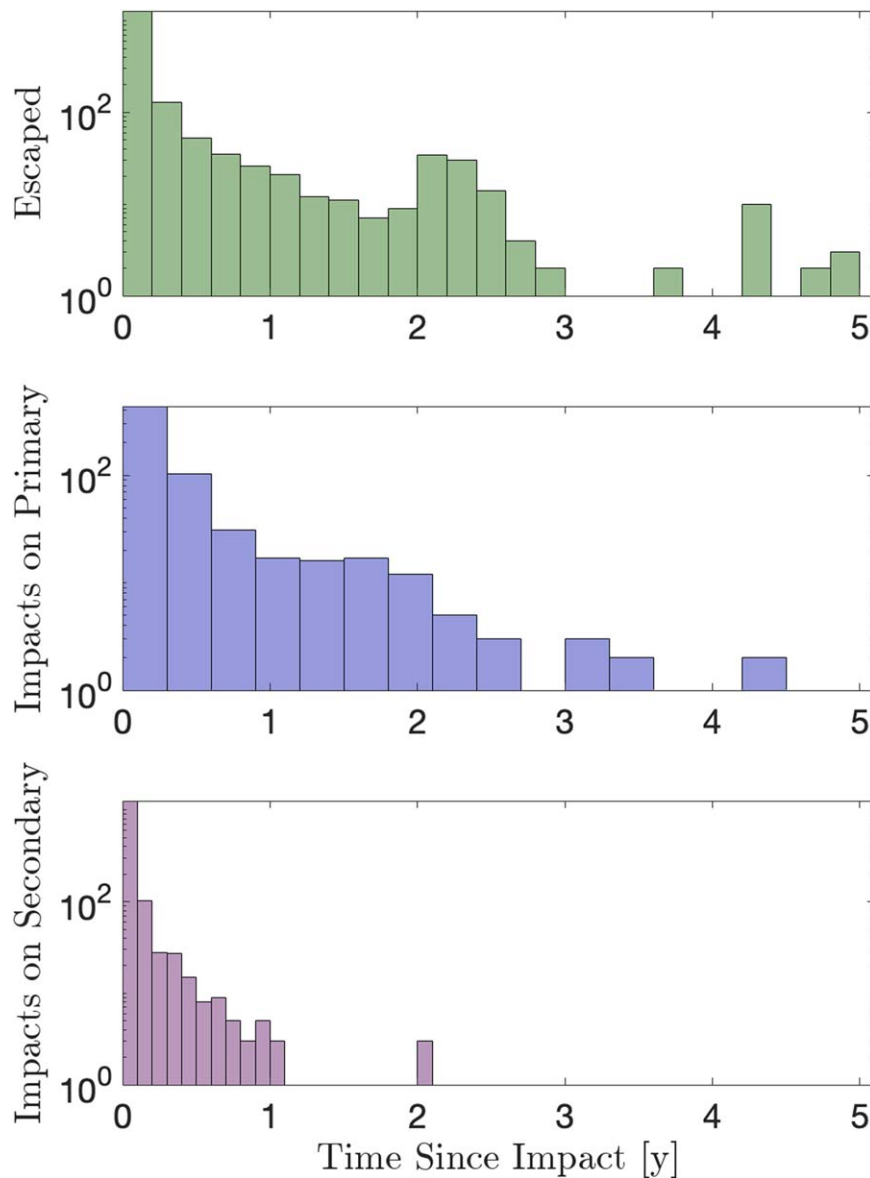


Figure 17. The fate of the ejecta over time, showing the number of particles that escape or impact with the primary or secondary. Impacts with the secondary generally only occur within the first year after the impact, while primary impacts or escapes can happen later.

this analysis will benefit from more detailed models of the shape and density of the asteroids.

6. Conclusion

The problem of particle dynamics around binary asteroids has been studied in depth in the literature. However, these studies do not consider the full dynamics within a perturbed system, as this problem becomes complex. We use an F2BP simulator, augmented to include the effects of solar gravity and radiation pressure, to study the dynamics of a particle under the influence of these forces. Our model can fully account for the effects of eccentricity, nonspherical gravity, and non-principal-axis rotation of the secondary. Within the Didymos binary asteroid after the DART impact, we find certain orbital regimes that are stable over several years. Notably, it is possible for ejecta created by the DART impact to last until the arrival of Hera, and these particles will likely be found in highly eccentric and inclined orbits around Didymos. The most likely

region for Hera to encounter these particles is just outside the orbit of Dimorphos. Circular orbits interior to Dimorphos remain stable despite the perturbation to the system. These orbits can be leveraged by Hera to study both asteroids while remaining relatively safe from any debris. However, any debris that does enter this regime will have large velocities relative to Hera.

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